On the **Stability** of Feature Selection in the Presence of **Correlations**

Konstantinos **Sechidis**, Konstantinos **Papangelou**, Sarah **Nogueira**, James **Weatherall** and Gavin **Brown**

MANCHESTER 1824
The University of Manchester
AstraZeneca

How much do you **trust your feature choices**?

How **reproducible** is your research result?

Stability

"the change in the selected feature subset caused by tiny changes in the training data"

Nogueira et al, JMLR 2018

Get your data. Select some features.
Drop a small number of examples and select again.
Did you select the same features?

If not, **how much can you trust** your original choices?

Stability (Nogueira et al, 2018) **can quantify this**.

I perturb my data 10 times:

$s_1 : \{x_1, x_3, x_6\}$, $s_6 : \{x_1, x_3, x_6\}$,
$s_2 : \{x_2, x_4, x_6\}$, $s_7 : \{x_2, x_4, x_6\}$,
$s_3 : \{x_2, x_3, x_6\}$, $s_8 : \{x_1, x_3, x_6\}$,
$s_4 : \{x_2, x_3, x_6\}$, $s_9 : \{x_1, x_4, x_6\}$,
$s_5 : \{x_1, x_4, x_6\}$, $s_{10} : \{x_2, x_4, x_6\}$.

$$\begin{array}{c} \begin{array}{cccccc} Z_1 & Z_2 & Z_3 & Z_4 & Z_5 & Z_6 \end{array} \\ \begin{bmatrix} 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix} \end{array}$$

$$\underbrace{\hat{\Phi}(\mathcal{Z})}_{\text{Stability measure}} = \frac{1}{M(M-1)} \sum_{i=1}^{M} \sum_{\substack{j=1 \\ j \neq i}}^{M} \underbrace{\phi(s_i, s_j)}_{\textit{Similarity}\text{ measure}} = 1 - \frac{\sum_f \frac{M}{M-1} \hat{p}_f (1 - \hat{p}_f)}{\bar{k}(1 - \frac{\bar{k}}{d})}$$

The above **selection matrix** has stability… 0.26

Not very stable. I cannot **trust** my original choices.

But what if I tell you x1 is **strongly correlated** to x2 ?

What if I have **domain knowledge** that x1 is the same thing as x2, but just a different name?

Look at the matrix again...

This paper proves :

- stability measures (any) will systematically **underestimate** in the presence of correlations
- leading to an **overly pessimistic view** of stability

This paper provides :

- a **correction** to incorporate domain knowledge on correlations, **giving the effective stability.**

Definition 2 (Effective Stability for Pairwise Feature Redundancy). *Given a matrix $\mathbb{C}$ specifying feature relationships, the effective stability is*

$$\hat{\Phi}_{\mathbb{C}}(\mathcal{Z}) = 1 - \frac{\sum_{f=1}^{d} \sum_{f'=1}^{d} c_{f,f'} \widehat{\text{cov}}(Z_f, Z_{f'})}{\sum_{f=1}^{d} \sum_{f'=1}^{d} c_{f,f'} \text{cov}(Z_f, Z_{f'} | H_0)} = 1 - \frac{\text{tr}(\mathbb{C}\mathbf{S})}{\text{tr}(\mathbb{C}\mathbf{\Sigma^0})}, \quad (5)$$

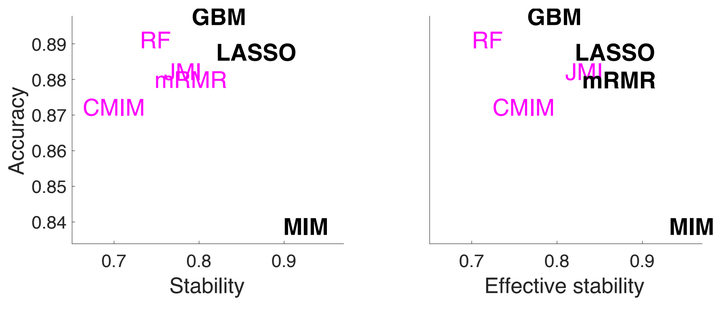
where $\mathbf{S}$ is an unbiased estimator of the variance-covariance matrix of $\mathcal{Z}$, i.e. $\mathbf{S}_{f,f'} = \widehat{\text{Cov}}(Z_f, Z_{f'}) = \frac{M}{M-1}(\hat{p}_{f,f'} - \hat{p}_f \hat{p}_{f'})$, $\forall\ f, f' \in \{1...d\}$, while $\mathbf{\Sigma^0}$ is the covariance matrix of $\mathcal{Z}$ under the null model of feature selection.

Empirical Study: Stability of Biomarker Selection

Rank	GBM	CMIM
1	EGFR expression (X_4)	EGFR mutation (X_2)
2	Disease stage (X_{10})	Serum ALP(X_{13})
3	WHO perform. status (X_1)	Blood leukocytes (X_{21})
4	Serum ALT(X_{12})	Serum ALT (X_{12})

Table 4: Stability and effective stability of GBM and CMIM in IPASS. The instability of CMIM is caused by variations within groups of semantically related biomarkers. When this is taken into account using $\hat{\Phi}_{\mathbb{C}}(\mathcal{Z})$ the method is deemed more stable than GBM.

	GBM		CMIM
Stability $\hat{\Phi}(\mathcal{Z})$	**0.87**	>	0.68
- within **Group A**	0.96		0.45
- within **Group B**	0.82		0.80
- within **Group C**	0.14		0.43
Effective stability $\hat{\Phi}_{\mathbb{C}}(\mathcal{Z})$	0.87	<	**0.91**



Accuracy vs Stability Pareto-optimality

Effective Stability identifies unstable solutions.

Nogueira et al, "On the Stability of Feature Selection Algorithms"
Journal of Machine Learning Research 18 (2018) 1-54.